CURRICULUM VITAE

Rajeev Kumar Shakya, PhD

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PhD – Indian Institute of Technology Kanpur UP
M.Tech, B.Tech (SGSITS INDORE MP)

Personal	Date of birth:	09 February 1979
Information:	Nationality:	Indian
	Marital Status:	Married

Personal Web Page:
https://sites.google.com/view/rajeevshakya/

Personal Linkedin Link:
https://www.linkedin.com/in/rajeevshakya/

Personal Skype ID:
rajeevgwl.shakya

Educations :

Name of Exam Passed	Board / University	Name of College	Year of Passing	Specialization
Ph.D	Indian Institute of Technology Kanpur India	Indian Institute of Technology Kanpur India	2015	Signal Processing Communication & Networks
M.Tech.	RGPV, Bhopal India	Shri G.S.I.T.S., Indore, MP, India	2007	Digital Techniques & Instrumentation
B.Tech.	RGPV, Bhopal India	Shri G.S.I.T.S., Indore, MP, India	2003	Electronics and Instrumentation Engineering
12th	M.P. Board Bhopal India	Carrier Convent School, Tansen Nagar, Gwalior MP India	1997	PCM Group
10th	M.P. Board Bhopal India	Govt. Boys Higher Sec. School, Kailaras Morena, MP India	1995	All subjects

Work Experiences: - Teaching, Research & Industry Experiences

S.No.	Institute Name	Post Held	Duration	Pay Scale
01	**Adama Science and Technology University, Adama, Ethiopia**	Associate Professor	Feb. 2020 to June 2025	as per international standard
	• To take theory as well as practical classes. • Guided B.Sc., M.Sc. and Ph.D. Students for their project work. • Educational Accreditation Coordinator • Coordinator of Electronics Projects Laboratory			
02	**Galgotias College of Engg. & Tech., Gr. Noida**	Associate Professor	Jan. 2015 to Dec. 2019	(Rs.15,600- 39,100 and Grad Pay Rs.9000)
	• To take theory as well as practical classes. • Guided B.Tech. Students for their project work. • NAAC, NIRF, NBA, Anti-Ragging Coordinator • Coordinator of Electronics Projects funded by MSME			
	Galgotias University, Gr. Noida	Assistant Professor	March 2014 to Dec. 2014	(Rs.15,600- 39,100 and Grad Pay Rs.8000)

03	• To take theory as well as practical classes. • Guiding B.Tech. Students for their project work. • In charge of the departmental library.			
04	**EE dept., Indian Institute of Technology (IIT), Kanpur**	Research Scholar	July 2008 to March 2014	PhD Scholarship
	• Designed and Simulated the Correlation-based Communication Protocols for Wireless Sensor Network applications. • Conducted Microprocessors and Microcontroller Lab for UG students • Involved in the assignment of tutorials on Computer Network theory and Applications. • Supervised Graduate and Post-graduate trainees for their projects.			
05	**EE dept., Indian Institute of Technology (IIT), Kanpur**	Project Associate	Sept.2007 to July 2008	(Rs.8,000–275-13,500)
	• Contributed in Test and analysis of Radio Frequency Identification (RFID) Systems using active and passive tags for different environmental conditions. • Developed software modules for collecting the data using RFID Systems. • Detailed report presented on comparison of two specific readers (a) Alien ALR-8800 and (b) ThingMagic Mercury4. Their performances have been evaluated in terms of Read Range, Threshold Power, Read Rate, Read Volume.			
06	**MP Automation Pvt. Ltd. Indore, MP**	System Testing Engineer	July 2003 to June 2005	(Rs.8,000–275-13,500)
	• Involved in the development of software products and technological solutions for various clients from industries. • Experience in PCB design, design and implementation of network systems (Zigbee, WiFi), design of actual hardware prototypes.			

Achievements & Workshops attended:

Awards and Honors:

1. Received Appreciation certification for the completion of the project entitled "IoT-based smart Management of waste in Hospital for efficient monitoring and managing waste life-cycle in order to reduce the impact of COVID-19 and future Pandemics", from the University-Industry Linkage and Technology Transfer Directorate, Adama Science and Technology University Ethiopia, June-2025.
2. Received Appreciation certification for completion of the project entitled Automated Detection and Classification of Cancerous White Blood Cells using CNN Deep Learning Model from the Research and Centers of Excellence Directorate, Adama Science and Technology University Ethiopia, April-2025.
3. Received appriciation as member in the development of curriculum for MSc in Mechatronics

Engineering, Dept. of Mechanical Engg., Adama Science and Technology University, Adama. Oct., 2023.

4. Received Travel Award from Dean Resource Planning and Generation (DRPG), IIT Kanpur to present a paper in IEEE conference TENCON 2012 held in Cebu City, Philippines, Nov-2012.
5. Received Travel Award from Dean Resource Planning and Generation (DRPG), IIT Kanpur to present a paper in IEEE conference WOCN 2012 held in Indore City, India, Sept-2012.
6. Received Research Fellowship for pursuing PhD program from IIT Kanpur.
7. Received Scholarship from MHRD for M.Tech Program.

Training and Workshop Attended:

1. Received C and C++ Certificates from CMC Limited, a joint venture of Tata Sons Ltd and Government of India.
2. Faculty Development Program (30 hours live online training) on " Applied AI & Deep Learning in Healthcare Data Analysis (MRI, CT SCAN, COVID-19, XRAY)", organized by EduxLabs (Esoir Business Solution LLP) from 9th Aug 2021 to 23rd Aug 2021.
3. Faculty Development Program on 'Advances in Communication and Networking', July 17-21, 2017, sponsored by A.K.T.U., Lucknow and organized by Galgotias Education Institutions, Greater Noida.
4. Faculty Development Program on 'Mobile Computing & Communication', July 3-7, 2017, sponsored by DST and organized by National Institute of Technology (NIT), Delhi.
5. Session Chairing on international conference on Green Computing & Internet of Things, 8-10 October 2015 at Galgotias Education Institutions, Greater Noida.
6. Workshop on 'Computational Intelligence: Theories, Applications and Future Directions', 14th July-2013 at EE dept., IITK.
7. Workshop on 'Introduction to Network Science' from 29th Aug-2011 to 2nd Sept-2011 at Mathematics Initiative section, IISc, Bangalore.
8. Workshop on 'Brainstorming Session on Health Monitoring of Automotives' from 9th to 10th Mar-2010 at EE dept., IITK.
9. National Workshop on 'Simulation of Networks using NS-2' (NWSN-2010) from 24th to 25th Sept-2010 at IEEE ISTAR student branch, Vallabh Nagar, Gujarat.
10. Short term programme on 'Principles and Practices in Radio Frequency Identifications (RFID)' from 10th to 15th Nov-2007 at IITK.

UG/PG/PhD Projects:

1. **Spatial Correlation-based Efficient Communication Protocols for Wireless Sensor Networks** (Ph.D thesis).

   The dissertation deals with the analysis, design of Correlation-based MAC and Routing Protocols for event-driven sensor network applications. A mathematical framework has been developed to capture the correlation between omni-directional sensor nodes. Using this framework, various key elements were investigated to enable energy-efficient methodologies that exploit the spatial correlation at both the Network and medium access control (MAC) layers.

2. **Design and Simulation of LVDT Signal Conditioner** (M.TECH., 4th Semester)

   Developed a LVDT Signal Conditioner System using new modified Costas Receiver Demodulator on Matlab based simulink environment and build an experimental test-bed using AD698 LVDT signal conditioner for evaluations.

3. **Car Model Analyzer System** (M.TECH., 3rd Semester)

   Implemented an analyzer for measuring accurately the speed of motor for driving a car. Designed prototype test-bed system using sensors and AT89C51 micro-controller, is an accurate and quick way of analyzing the speed of car directly.

4. **Temperature and Flow Control System** (B.E., 8th Semester)

   Thesis deals with design and implementation on temperature and flow control system for industrial applications. System implementation and analysis done had been undergone and successfully completed in PLAVT lab at Centre for Advanced Technology (CAT) INDORE. It is a unit of Department of Atomic Energy, engaged in R and D in non-nuclear front-line research areas of Lasers, Particle Accelerators and related technologies.

List of Projects (Fully Funded):

1. Entitled as "Automatic Detection and Classification of Cancerous White Blood Cells Using CNN Deep Learning Model", Fully Funded Research Project, Completed duration from Aug-2021 to 2024, Achievements: One Research Article Published, [Completed].
2. Entitled as "IoT-based Smart Management of waste in Hospital for efficient monitoring, managing waste life-cycle in order to reduce the impact of COVID-19 & Future Pandemics", Fully Funded Technology Transfer Project for the duration Feb-2022 to 2024, ASTU, Ethiopia.Achievements: One Indian Patent published, one Commercialization startup [Completed].

3. Entitled as "Design and Implementation of Control and Navigation System for Agricultural Pesticides/Fertilizers Spraying Robotic Platform for improvement of the Crop Production in Ethiopia", Fully Funded Technology Transfer Project for the duration March-2025 to 2026, [Ongoing].

List of Publications (Published):

1. K Saminathan, L. Perusal, F. H. Sharon, and Rajeev K Shakya, "Multicast On-Route Cluster Propagation to Detect Network Intrusion Detection Systems on MANET using Deep Operator Neural Networks", International Journal of Expert Systems with Applications, Dec.2024, no. 125864, https://doi.org/10.1016/j.eswa.2024.125864, SCI, IF7.5.

2. G. Sakya, V. Yadav, S. Shukla, A. Gupta, and Rajeev K. Shakya, "An Intelligent IoT Based Landfill Fire Prediction and Prevention System", Wireless Personal Communication, 139, pp. 1837-1861, 2024, DOI: https://doi.org/10.1007/s11277-024-11702-2, SCI, IF6.5.

3. Stephen, V. M. Sundari, B. Thayumanavar, Rajeev Kumar Shakya, & E. Muthukumar, "Technology And Its Role In Shaping The Future Of Work Life Balance", Int. Journal of Educational Administration: Theory and Practice, vol. 30, no. 5, 1045–1053. https://doi.org/10.53555/kuey.v30i5.3005, 2024, SCOPUS.

4. B. Nigussie, Rajeev K Shakya, J. Shaikh, E.T. Ware, "Tabular Deep Learning based light-weight Intrusion detection System for VANET system", Internal journal of Technische Sicherheit, Vol. 24, no. 9, 2024. SCOPUS

5. P. Mamoria, G.G. Gebrealif, D. J. Gelmecha, S. Chura and R. K. Shakya "Overview Of Channel Coding Schemes and Its Comparative Analysis For 5G Wireless Networks" International Journal of Membrane Science and Technology, vol. 10, no. 5, 2023, pp.1207-1219.SCOPUS

6. P. Mamoria, Rajeev K Shakya, "A IoT based Health Monitoring System for pregnant rural women", Inter journal of Technische Sicherheit, Vol. 23, no. 12, pp. 275-281, 2023.SCOPUS

7. Rajeev K Shakya, E. T, Ware, and Teklu Urges, Susceptible-infectious Epidemiological Based Model With Spatial Correlation For Combating Virus Spreading In Wireless Sensor Networks, Internal journal of Technische Sicherheit, vol.23, no.11, pp. 145-152, 2023. SCOPUS

8. T. Gopi Krishna, Rajeev K. Shakya, M. Abdeldaieem Mahboub, Water Cycle Approach With Relevance Vector Ml For Brain Tumor Detection And Enhancement From Mri Images, Internal journal of Technische Sicherheit, vol.23, no.11, pp. 248-261, 2023. SCOPUS

9. Y. Abebaw, Rajeev Kumar Shakya, D. J. Gelmecha, and E. T. Ware, "Spectral Efficiency Analysis for Uplink Multi Cell Massive MIMO Cellular Communication System under Fading Channels", Journal of Electrical and Computer Engineering, vol. 2023, ID 6623938, 16 pages, 29-Sept-2023.SCOPUS

10. Rajeev K. Shakya, K. Devarajan, S.M. Samuel, S.Kumar, and T. Swathi, "The Role of Artificial Intelligence and Machine Learning in Supply Chain Optimization: A Systematic Review" International Journal of Harbin Engineering university, pp.1785-1793, vol. 44, No. 7, July- 2023.SCOPUS

11. B.Haldar, Rajeev K. Shakya, S. Kumar, N. Jaiswal, and A. Singh, "Ethical Considerations in AI-MI Adoption in Financial Institutions: A literature Review" International Journal of Harbin Engineering university, pp. 1493-1503, vol. 44, No. 7, July-2023.SCOPUS

12. M. Agarwal, Rajeev K. Shakya, A.L.S, L.Devi, and N. Raja, "Renewable Energy Sources and Their Potential for Sustainable Development: A comprehensive Review," International Journal of Harbin Engineering university, pp.1472-1481, vol. 44, No. 7, July-2023.SCOPUS

13. Hise Teferi Dumari, Demmissie J Gelmecha, Rajeev K Shakya and Ram S Singh, "BER and PSD improvement at FBMC with Higher Order QAM using Hermite Filter for 5G wireless Communication and Beyond", Journal of Electrical and Computer Engineering, 16 pages, https://doi.org/10.1155/2023/7232488, 21-Dec. 2022.SCOPUS

14. Rajeev K. Shakya, Tadesse Hailu Ayane, Feyissa Debo Diba and P. Mamoria, "SEIRS Epidemic model with spatial correlation for analyzing dynamic of virus spreading in event-driven wireless sensor networks," International Journal of System Assurance Engineering and Management, vol. 13, No. 2, pp. 752-760, April, 2021.SCI

15. S.K. Chaudhary, Rajeev K. Shakya, and A. Kumar, "SIR epidemic model derived from spatial correlation for worm propagation in event-driven wireless sensor network," International Journal of Innovative Technology and Exploring Engineering, vol.-9, no.-3, pp. 1-15, Jan-2020.

16. Rajeev K. Shakya, K. Rana, A. Gaurav, P. Mamoria and P.K.Srivastava, "Stability Analysis of Epidemic Modeling Based on Spatial Correlation for Wireless Sensor Networks," International Journal of Wireless Personal Communication, Springer, vol.-106, no.-3, pp. 1-15, June-2019.SCI

17. Rajeev K. Shakya, Y. N. Singh, Nishchal Verma, "Generic Correlation Model for Wireless Sensor Network Applications," Journal IEEE/IET Wireless Sensor Systems journal, vol. 3, no. 4, pp. 266-276, Dec 2013 (doi:10.1049/iet-wss-2012.0094).

18. Rajeev K. Shakya, Yatindra N. Singh, Nishchal Verma, "A Correlation Model for MAC Protocols in Event-Driven Wireless Sensor Networks," in Proc. of the 2012 IEEE Region 10 Conference, IEEE TENCON-2012, Cebu, Philippines, Nov 19-22, pp. 1–6, 2012.

19. Rajeev K. Shakya, Yatindra N. Singh, Nishchal Verma, "A Novel Spatial Correlation Model for Wireless Sensor Network Applications," In Proc. of the Ninth IEEE International Conference on

Wireless and Optical Communications Networks (IEEE WOCN-2012), Indore, INDIA, Sept., 20-22, pp. 1-6, 2012.

20. Satyam Agarwal, Rajeev K. Shakya, Yatindra N. Singh, and Amitabha Roy, "DSAT-MAC: A Dynamic Slot Allocation TDMA MAC protocol for Cognitive Radio Networks," In Proc. of the Ninth IEEE International Conference on Wireless and Optical Communications Networks (IEEE WOCN-2012), Indore, INDIA, Sept., 20-22, pp. 1-6, 2012.
21. Rajiv Tripathi, Rajeev K. Shakya, Nishchal K. Verma and Yatindra N. Singh, "Localized detection of bottleneck nodes and quantification of criticality in a wireless sensor networks," 2010 IEEE Region 8 International Conference on Computational Technologies in Electrical and Electronics Engineering (SIBIRCON 2010), Russia, pp. 326-328, 2010.
22. Pushpa Mamoria, Rajeev K. Shakya and M. Hima Bindu, "A Backoff Analysis to mitigate the Correlated Contention for Event-driven Wireless Sensor Networks," National Conference on Current Trends in Computing, NCCTC-2014, North Orissa University, Orissa, pp. 1-6, 2014.

List of Book Chapters

1. Rajeev Kumar Shakya, T . Abebaw, D. J. Gelmecha, and E. T. Ware, Book Chapter entitled " Improved Spectral Efficiency Analysis for Uplink Multi Cell Massive MIMO Cellular Communication Systems Using MMSE Channel Estimation and ZF Uplink Combining", Inter. Book Theory and Applications of Engineering Research Vol.2, DOI:https://doi.org/10.9734/bpi/taer/v2/8384A, 01–Jan-2024.
2. P. Mamoria, Rajeev Kumar Shakya and J. Agrwal, Book Chapter entitled "Broader Implication of Artificial Intelligence on business management: Opportunity and Challenges", in the book entitled as 'Emerging Frontiers: Interdisciplinary Perspectives on Commerce, Economics and Management', First Edition, Infinity Publications, chapter-25, pp.152-159, ISBN:978-93-89476-39-2, DOI: 10.25215/9389476399, July, 2023, https://redshine.co.in/product/9789389476392/.
3. P. Mamoria and Rajeev Kumar Shakya, Book Chapter-11 entitled "The Challenges and Importance of the Green Economy: A Global Survey", in the book entitled as 'Emerging Frontiers: Interdisciplinary Perspectives on Commerce, Economics and Management', First Edition, Infinity Publications, chapter-11, pp. 66-72, ISBN:978-93-89476-39-2, July, 2023, https://redshine.co.in/product/9789389476392/.
4. Rajeev Kumar Shakya and Pushpa Mamoria, book chapter entitled "MODELING SIRD MODEL WITH SPATIAL CORRELATION FOR WIRELESS SENSOR NETWORK APPLICATIONS", Emerging Issues In Commerce, Economics, Information Technology And Management (Vol – II), publisher

Redshine Publication, DOI: 10.25215/9789390937318, (https://redshine.co.in/product/978-93-90937-31-8/), 2021.

List of Patent Filed/Published/Granted

1. P.Mamoria, R.K.Shakya, "SMART WALLET WITH FINGERPRINT LOCK AND GPS TRACKER", filed with application no 202411066344, 02-sept-24.
2. D. J. Gelmecha, R.K.Shakya, E.T. Ware, and P.Mamoria, "Intelligent waste containers design and wireless communication networks and control of waste containers, managing status information thereof", Ethiopian Patent Filed on dated 22 nov. 2024.
3. D. J. Gelmecha, R.K.Shakya, G.Dengia, and T.Hailu, "SMART DUSTBINS FOR MONITORING HOSPITAL WASTE MANAGEMENT", Application no. 426083-001, filed and published dated July 2024.
4. L. Singh, S. S. Bisht, M. S. Choudhry, A. Alam, G. Sakya, S. Sinha, and Rajeev Kumar "An IOT based Healthcare System for Predicting Severity Prognosis in Heart Patients using Machine Learning", Indian Patent Granted, dated May 2023.
5. Patent entitled "AN AI-POWERED MENTAL HEALTH DIAGNOSIS AND ASSESSMENT SYSTEM AND METHOD THEREOF", IPO India publisher, Application no.:202311030931, 2023.
6. Patent entitled "Vedic-Mathematics based Effective High- Speed and Low Power Multiplier Architecture using for DSP Application", IPO India publisher, Application no.:202241003733, 2023.
7. Patent entitled "Intelligent Printer: Remove the Printed Text (color, Black) and Print Again", Filing Date-09-05-2021, IPO Australian, publisher, date-09-05-2021.

Extracurricular activities:

International Conference (Coordinated/Organized):

1. Organized first IEEE Uttar Pradesh Conference, International Conference on *"Energy, Economics and Environment" (1st UPCON2015)*, GCET Greater Noida" March 26-28, 2015.

National Conference (Coordinated/Organized):

1. Organized an IEEE sponsored national conference on *"Energy, Power and Intelligent Control Systems (EPICS-2014)"*, GCET Greater Noida March 28-29, 2014.

National Seminar (Coordinated/Organized):

1. *"Empowering young Entreprenuers in India - Promoting Ambitition & Innovation"* in association with Indian Institute of Technology Kharapur, GCET Greater Noida, Sept. 15-16 2016.

Short Term Training Program (Organized):

1. Organized a short term training program on "*DCS and PLC"* at GCET Gr. Noida (Aug.12, 2016 to Aug.14, 2016) by CETPA InfoTech Pvt. Noida.
2. Organized a short term training program on *MATLAB tools and its implementation* at GCET Gr. Noida (Sept.25, 2016 to Sept.28, 2016) by CETPA InfoTech Pvt. Noida.

Workshop (Organized):

1. Organized One-day workshop on *"incubation and startup"* on 17th March 2016 in support of Ministry of MSME (Micro, Small & Medium Enterprises Govt. of India at GCET, Greater Noida.
2. Organized Two-days workshop in collaboration with IIT Delhi on the topic *"Internet of Things – IOT"* (1-2 March 2016) at GCET, Greater Noida.
3. Organized One day workshop in technical collaboration with CETPA Pvt. Ltd Noida on topic on *"Industrial Automation in context with PANASONIC PLC"* (Feb. 20, 2015).

Laboratories Setup

Microprocessors Lab, Network Lab, Research & Development Lab

Reviewer Member of International Journals:

Computer applications "Taylor & Francis" and Joural of Computer Networks "Elsevier", International Journal of System Assurance Engineering and Management "Springer", Wireless Personal Area Communication "Springer".

Subject Taught:

Microprocessors and Its Applications	Data Communication Networks and Security
Introduction to Wireless Communication	Introduction to Satellite Communication
Electrical Measurement & Measuring Instrument	Embedded and Real-Time System
Advanced Microprocessors and Microcontrollers	Advanced Real Time Embedded System
Analytical and Computational Methods in Engineering	Radio Frequency Engineering
Electrical and Electronics Engineering Materials	Neural Networks Fuzzy Logic and Control

Area of Research:

NS-2/NS-3 simulator Freelancer trainer with various institutions on Wired/Wireless Protocols, MAC layer, Routing/Network Layer, Queue Management, Quality of Service (QoS), Multiple Channels, Cognitive

Radio/Vehicular Radio Network, MIMO-OFDM, 4G/5G NR, and Machine Learning in Wireless Communication.

Strength: Perfection in work, Leadership, Strong will power, Quick learner with Positive attitude.

I hereby declare that all the aforesaid information is correct and complete to the best of my knowledge.

(Dr. Rajeev K Shakya)